

Density Functional Theory

We will use the **non-relativistic many-body Hamiltonian** for electrons

$$\hat{H} = -\frac{\hbar^2}{2m_e} \sum_i \nabla_i^2 - \sum_{i \in \mathbb{I}} \frac{Z_i e^2}{|r_i - R_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|r_i - r_j|} - \frac{\hbar^2}{2} \sum_{\mathbb{I}} \frac{\nabla_{\mathbb{I}}^2}{M_{\mathbb{I}}} + \frac{1}{2} \sum_{\mathbb{I} \neq \mathbb{J}} \frac{Z_{\mathbb{I}} Z_{\mathbb{J}} e^2}{|R_{\mathbb{I}} - R_{\mathbb{J}}|}$$

kinetic energy of electrons, nuclear potential, potential of nuclei, electron-electron interaction, kinetic energy of nuclei, nuclear interaction

We will be interested in calculating **observables**

$$O = \langle \hat{O} \rangle = \frac{1}{Z} \sum_{\alpha} e^{-\beta(E_{\alpha} - \mu N_{\alpha})} \langle \Psi_{\alpha} | \hat{O} | \Psi_{\alpha} \rangle = \frac{1}{Z} \text{Tr}(e^{-\beta(\hat{H} - \mu \hat{N})} \hat{O})$$

many body state

At $T=0$, N fixed,

$$O = \int \dots \int dx_1 \dots dx_N \Psi^*(x_1, \dots, x_N) \sum O(x_i, x_j, \dots) \Psi(x_1, \dots, x_N)$$

For each $\hat{H}$, i.e. for each V_{ext} , we can view O as a functional on the space of states. We move from viewing $O = O[\Psi]$ as a simple functional of complicated wave functions to viewing $O = \tilde{O}[V_{\text{ext}}]$ as a complicated functional of a simple function.

Note that if $O(x_i, x_j, \dots) = O_i(x_i)$ then

$$O_i = \int dx_1 O_i(x_1) N \int dx_2 \dots dx_N \Psi^*(x_1, \dots, x_N) \Psi(x_1, \dots, x_N) = \int dx_1 O_i(x_1) \cdot n(x_1)$$

one-body density

For $O(x_i, x_j) = \frac{e^2}{|r_i - r_j|}$, the Coulomb potential, we similarly get

$$O = V_{ee} = \iint dx_1 dx_2 \frac{e^2}{|r_1 - r_2|} \frac{N(N-1)}{2} \int dx_3 \dots dx_N \Psi^*(x_1, \dots, x_N) \Psi(x_1, \dots, x_N) = \iint dx_1 dx_2 \frac{1}{2} \frac{e^2}{|r_1 - r_2|} n^{(2)}(x_1, x_2)$$

electron-electron interaction, two-body density

For $O(x_i, x_j, \dots) = -\nabla_i^2/2$, we denote the right hand Ψ 's input by x_i' and ∇_i by ∇_i' , to remember that it only acts on the right hand Ψ , and get

$$E_{\text{kin}} = -\int \dots \int dx_1 \dots dx_N \Psi^*(x_1, \dots, x_N) \sum \frac{\nabla_i^2}{2} \Psi(x_1, \dots, x_N) = -\int dx_1 \left(\frac{\nabla_1^2}{2} N \int \dots \int dx_2 \dots dx_N \Psi^*(x_1, \dots, x_N) \Psi(x_1, \dots, x_N) \right)_{x_1=x_1'}$$

density matrix

Examples of functionals

- $F[f] = \int dx g(f(x))$ (local)
- $F[f_1, f_2] = \int dx dx' W(x, x') f_1(x) f_2(x')$ (nonlocal)

Let $F = C^0(\mathbb{R})$. A function $\phi: F \rightarrow F$ is **local** if $\exists n \geq 0, \bar{\phi}: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ for which $\phi[f]: x \mapsto \bar{\phi}(x, f(x), \dots, f^{(n)}(x))$

for all $f \in F$. A functional $\Phi: F \rightarrow \mathbb{R}$ is called **local** if it can be written as

$$\Phi: f \mapsto \int dx \phi[f](x)$$

for a local function $\phi: F \rightarrow F$.

We define the following density functionals:

$$E_{\text{ext}}[n] = \int dx n(r) V_{\text{ext}}(r)$$

$$E_H[n] = \frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|}$$

$$T^{\text{TF}}[n] = C \int dx n^{\frac{5}{2}}(r)$$

$$T^{\text{W}}[n] = \frac{1}{8} \int dx \frac{|\nabla n(r)|^2}{n(r)}$$

where n is the one-body density from above.

Recall that $H = T + V_{ee} + V_{\text{ext}} = -\frac{1}{2} \sum \nabla_i^2 + \sum_{i \neq j} \frac{1}{|r_i - r_j|} + \sum V_{\text{ext}}(r_i)$. Note that $O = \langle \Psi | \hat{O} | \Psi \rangle$. We will assume that V_{ext} determines a unique Ψ , its ground state, i.e. H does not have degenerate states.

Theorem (Hohenberg-Kohn): There is a bijection between the set of V_{ext} (without degenerate states) and wave functions Ψ , and density functions $n = n(r)$ that are V -representable.

Assume $H_1 |\Psi_1\rangle = E_1 |\Psi_1\rangle, H_2 |\Psi_2\rangle = E_2 |\Psi_2\rangle$. Then $(V_{\text{ext}}^{(1)} - V_{\text{ext}}^{(2)}) |\Psi_2\rangle = (E_1 - E_2) |\Psi_2\rangle$, i.e. $V_{\text{ext}}^{(1)}(r) = V_{\text{ext}}^{(2)}(r) + E_1 - E_2 = V_{\text{ext}}^{(2)}(r) + \text{const.}$

Let $H_1 |\Psi_1\rangle = E_1 |\Psi_1\rangle$, with Ψ_1 the ground state. Then

$$E_1 = \langle \Psi_1 | H_1 | \Psi_1 \rangle = \langle \Psi_2 | H_1 | \Psi_2 \rangle = \langle \Psi_2 | H_1 - H_2 | \Psi_2 \rangle + \langle \Psi_2 | H_2 | \Psi_2 \rangle = E_2 + \langle \Psi_2 | V_{\text{ext}}^{(1)} - V_{\text{ext}}^{(2)} | \Psi_2 \rangle = E_2 + \int dx (V_{\text{ext}}^{(1)}(r) - V_{\text{ext}}^{(2)}(r)) n_2(r)$$

Similarly, $E_2 \leq E_1 + \int dx (V_{\text{ext}}^{(2)}(r) - V_{\text{ext}}^{(1)}(r)) n_1(r)$. If $n_1 = n_2$, we get $E_1 + E_2 \leq E_1 + E_2$.

Hence, we have $O = O[\Psi] = O[V_{\text{ext}}] = O[n]$ is a functional depending on the electron density.

Corollary: Let $E[\Psi] = \langle \Psi | H | \Psi \rangle$. Then $\min_{\Psi} E[\Psi] = \min_{n \text{ valid}} E[n]$.

We call $F_{\text{HK}}[n] = T[n] + E_{\text{ee}}[n]$ the **Hohenberg-Kohn**, or **universal**, **functional**. Then $F_{\text{HK}} = E[n] - \int dx V_{\text{ext}}(r) n(r)$. It does not have an explicit description.

Recall that the ground state density is an integral of the ground state wave function:

$$n(x_i) = N \int dx_2 \dots dx_N \Psi^*(x_1, \dots, x_N) \Psi(x_1, \dots, x_N)$$

This is too hard to calculate already because Ψ has interactions. We would like to build an auxiliary system of non-interacting particles with the same density. The goal is to replace

$$\left(-\sum \frac{\nabla_i^2}{2} + \sum V_{\text{ext}}(r_i) + \frac{1}{2} \sum_{i \neq j} \frac{1}{|r_i - r_j|} \right) \Psi(r_1, \dots, r_N) = E \Psi(r_1, \dots, r_N)$$

$$\text{by } \left(-\sum \frac{\nabla_i^2}{2} + \sum V_{\text{aux}}(r_i) \right) \Psi_{\text{aux}}(r_1, \dots, r_N) = E_{\text{aux}} \Psi_{\text{aux}}(r_1, \dots, r_N)$$

Then $n(r) = \sum_{\text{occupied}} |\Psi_e^{\text{aux}}(r)|^2$.

We will assume that such Ψ^{aux} is representable by some V_{aux} .

We have

$$E[n] = T[n] + \int dx n(r) V_{\text{ext}}(r) + E_{\text{ee}}[n] = F_{\text{HK}}[n] + \int dx n(r) V_{\text{ext}}(r) + T_S[n] - T_S[n] + \frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|} - \frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|}$$

$$= T_S[n] + \int dx n(r) V_{\text{ext}}(r) + \left(\frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|} \right) + F_{\text{HK}}[n] - T_S[n] - \frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|}$$

$E^{\text{H}}[n]$
 $E_{\text{xc}}[n]$

We have $E_{\text{xc}}[n] = T[n] - T_S[n] + E_{\text{ee}}[n] - E^{\text{H}}[n]$.

$\leftarrow \ll T[n]$ if they are approximations

Theorem (Kohn-Sham equations): Ψ_i minimises $E[n]$ s.t. $\int \Psi_i^*(r) \Psi_i(r) dr = 1$ iff

$$\left(-\frac{\nabla^2}{2} + V_{\text{ext}}(r) + \int dx' \frac{n(r')}{|r - r'|} + \frac{\delta E_{\text{xc}}[n]}{\delta n(r)} \right) \Psi_i(r) = \lambda_i \Psi_i(r)$$

$V^{\text{H}}(n, r) = V_{\text{ee}}^{\text{H}}(n, r)$
 $V^{\text{eff}}(r) \text{ or } V_{\text{KS}}[n, r]$

The state space has a basis of eigenvectors for KS

To see this, calculate $\frac{\delta}{\delta \Psi_i} (E[n] - \lambda \int \Psi_i^*(r) \Psi_i(r) dr)$ for each term. Use that $\frac{\delta}{\delta \Psi_i} = \frac{\delta n}{\delta \Psi_i} \frac{\delta}{\delta n} = \Psi_i \frac{\delta}{\delta n}$.

From the KS-equations, we get

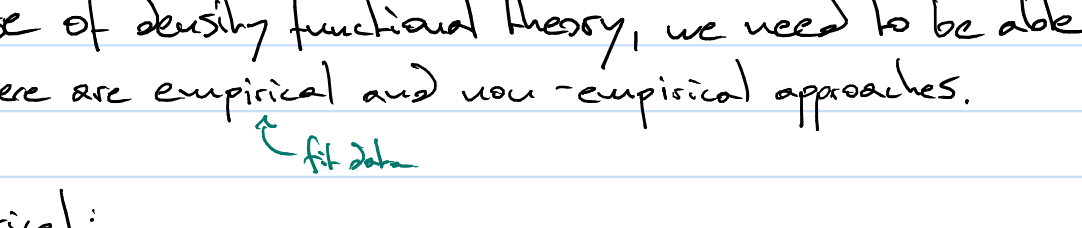
$$\sum \int dx \Psi_i^*(r) K_S \Psi_i(r) = \sum \int dx \Psi_i^*(r) \lambda_i \Psi_i(r)$$

i.e.

$$T_S[n] = \sum \lambda_i - \int dx n(r) V_{\text{ext}}(r) - \int dx \frac{n(r) n(r')}{|r - r'|} - \int dx n(r) v_{\text{xc}}[n, r]$$

Substituting, we get

$$E[n] = \sum \lambda_i - \frac{1}{2} \int dx dx' \frac{n(r) n(r')}{|r - r'|} + E_{\text{xc}}[n] - \int dx n(r) v_{\text{xc}}[n, r]$$



Algorithm (Kohn-Sham):

- $V_{\text{KS}} \leftarrow V_{\text{ext}}$
- n_{KS} is the density of Ψ s.t. $K_S \Psi = \lambda_0 \Psi$
- $V_{\text{KS}} \leftarrow V_{\text{KS}} + \mu(n_{\text{KS}} - n^0)$, where μ, p are stability parameters

We now call the **universal functional**

$$F[n] = T_S[n] + E_H[n] + E_{\text{xc}}[n]$$

To make use of density functional theory, we need to be able to approximate $E_{\text{xc}}[n]$. There are empirical and non-empirical approaches.

Non-empirical:

- local** - Local density approximation assume homogeneous gases locally, so functionals become functions
- semilocal** - Generalised gradient approximation in addition to above, use a function of n and ∇n
- semilocal** - Meta-GGA the function is a function of $n, \nabla n$, and τ , where $\tau = \tau(r)$ is the kinetic energy density $\frac{1}{2} \sum |\nabla \Psi_i(r)|^2$
- nonlocal** - Hybrid mix exact exchange-correlation with any of the above
- nonlocal** - RPA-like

Self-consistent field approach (SCF)

$$\left[-\frac{1}{2} \nabla^2 + V_{\text{ext}}(r) + V_H[n, r] + V_{\text{xc}}[n, r] \right] \Psi_i(r) = \epsilon_i \Psi_i(r)$$

$$V_H[n, r] \leftarrow V_{\text{ee}}^{\text{H}}[n, r] \quad n(r) = \sum |\Psi_i(r)|^2$$